EXPLORING AUTOENCODERS FOR NON-LINEAR DIMENSIONALITY REDUCTION AND IMAGE RECONSTRUCTION ON MNIST

GITHUB: HTTPS://GITHUB.COM/VINAYGEDDAM/MACHINE-LEARNING-ASSIGNMENT

Introduction

Contemporary machine-learning systems are often applied to high-dimensional data, especially images in which a single input may involve thousands of dimensions or features. The complexity involved can be overseen by deep learning models but at an extremely high computational cost and in very high-dimension space, there are often redundancies which limit efficiency. The aim of dimensionality reduction techniques is to transform data into a smaller representation that does not lose the crucial information.

Autoencoders are a neural network-based method that supports this. Unlike techniques such as Principal Component Analysis (PCA), which rely on linear transformations, autoencoders can model complex non-linear relationships. With this technique, they can capture richer structures, especially for image datasets with non-linear pixel dependencies.

In this tutorial, we will see how convolutional autoencoders can be utilized to compress and reconstruct images using MNIST. The analysis of the latent space structure, the patterns of reconstruction error, and the extension of the model to a noisy input go beyond reconstruction alone. Through the experiments in the tutorial, it highlights how the autoencoder learns this representation that is useful for dimensionality reduction and data denoising.

Conceptual Foundations

→**Representation Learning:** Representation learning is central to the functioning of the autoencoder. Neural networks learn the structures of data automatically rather than the more traditional wised handcrafted features. In terms of images, this means identifying patterns like edges, curves, and shapes that define objects.

Since autoencoders learn to reproduce their inputs, they force the model to compress information to a smaller latent representation. As a result of this compression, the network reduces to only relevant features.

→**Architecture of Autoencoders:** An autoencoder consists of two parts.

- The encoder maps the input into a lower dimensional latent space.
- The decoder creates the input from the compressed representation.

The goal of training is to minimise the reconstruction loss which makes the output like the input.

→**Non-Linearity Advantage:** PCA and other traditional dimensionality reduction methods assume linear relationships between variables. Nevertheless, there are often non-linear patterns in image data by employing non-linear activation functions, autoencoders can overcome this hurdle and model complex transformations.

These are quite useful for heavy image compression and reconstruction.

→**Latent Space Interpretation:** The latent space is often more than simply a compressed and lower dimensional representation of the input. In this space, similar inputs will be mapped to near points to achieve clustering and visualisation.

By analysing the structure of the latent space, we understand how the model has interpreted the data.

→**Robust Feature Learning through Denoising:** Denoising autoencoders introduce noise into input data during training and require the model to reconstruct the original clean version. This forces the network to learn robust features that are invariant to noise, improving generalisation.

Experimental Design

→**Dataset Description:** The foundation of this tutorial relies on the MNIST dataset. It contains grayscale images of handwritten digits which have the resolution of 28×28. Furthermore, it has around 70,000 such images. The dataset has a simple structure which is easy to evaluate. Hence, it is widely used in representation learning.

→**Model Configuration:** To leverage the spatial relationships, a convolutional autoencoder architecture was implemented. The encoder uses convolutional and pooling layers to gradually reduce dimensionality, while the decoder uses up sampling layers to project the image back.

The bottleneck layer is the compressed latent space where the input dimension is highly reduced.

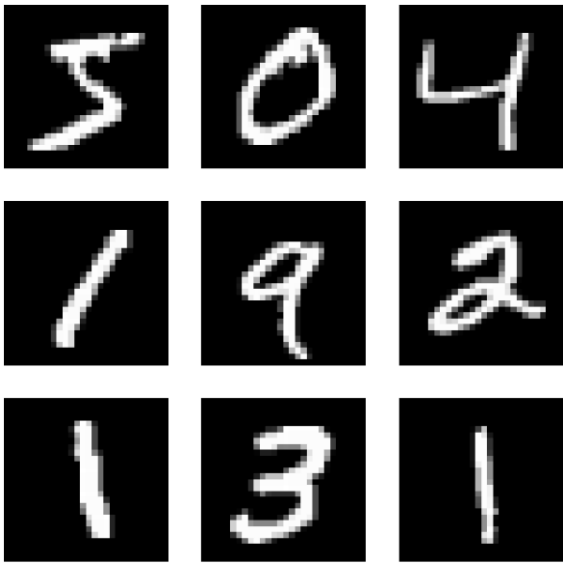
→**Training Strategy:** The model was trained using the Adam optimizer and binary cross-entropy loss. A small number of epochs was sufficient to demonstrate convergence and reconstruction capability.

To extend the analysis, additional experiments were conducted:

- Latent space visualisation using t-SNE.
- Reconstruction comparison between original and output images
- Denoising autoencoder with artificially corrupted inputs
- Reconstruction error distribution analysis

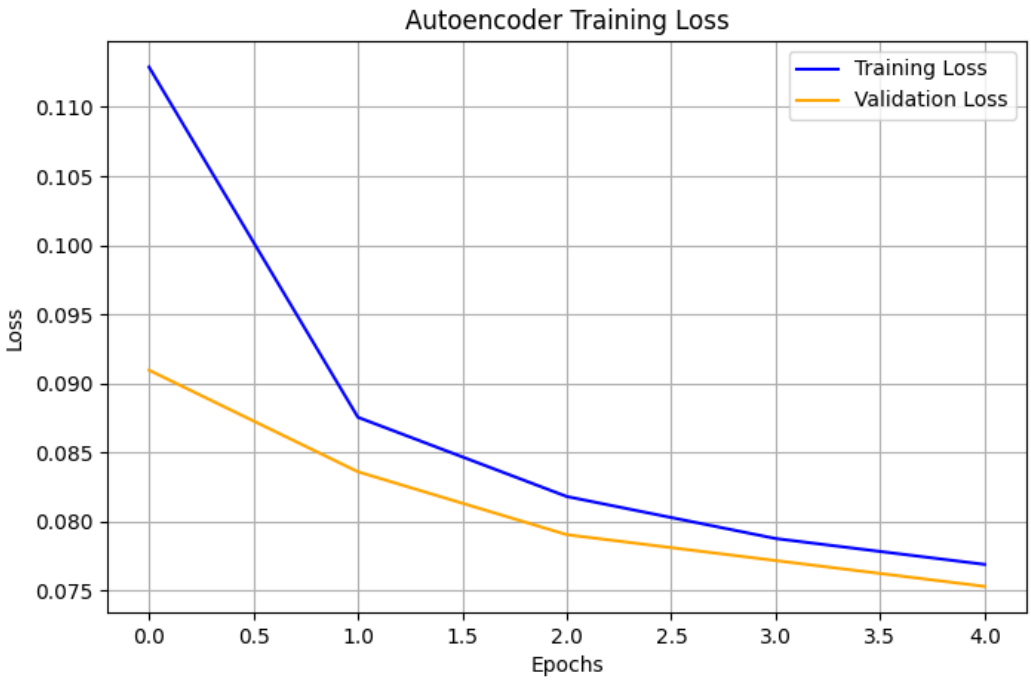
Results and Interpretation

→**Initial Data Exploration:** Visual inspection of MNIST samples highlights variability in handwriting styles. Differences in stroke thickness, orientation, and shape introduce complexity, requiring the model to capture essential structural features rather than memorising pixel values.



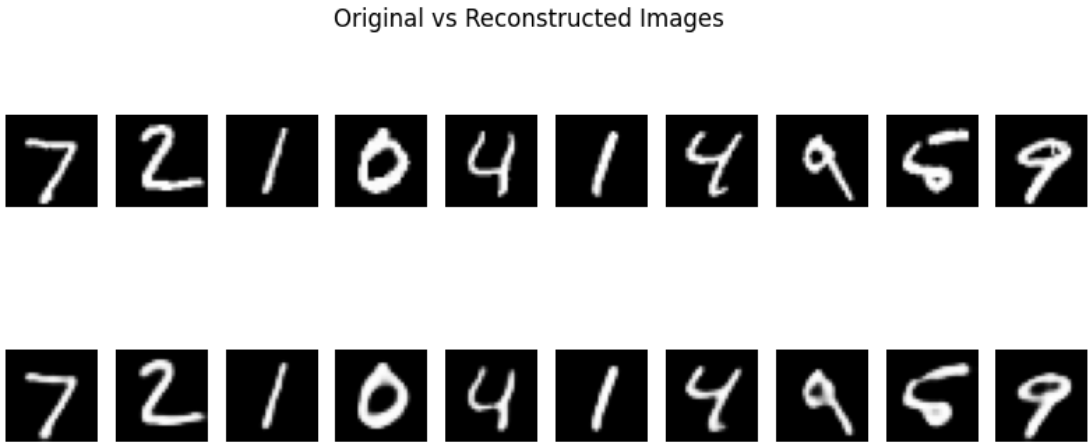
→**Training Dynamics:** The decline in the training loss curve over epochs is a good sign. It shows that the model can learn required features to reconstruct the input images. The similarity between training and validation loss shows that the model generalises well and does not overfit the data.

We see that the architecture is suitable for the task since the latent space captures the information we want.



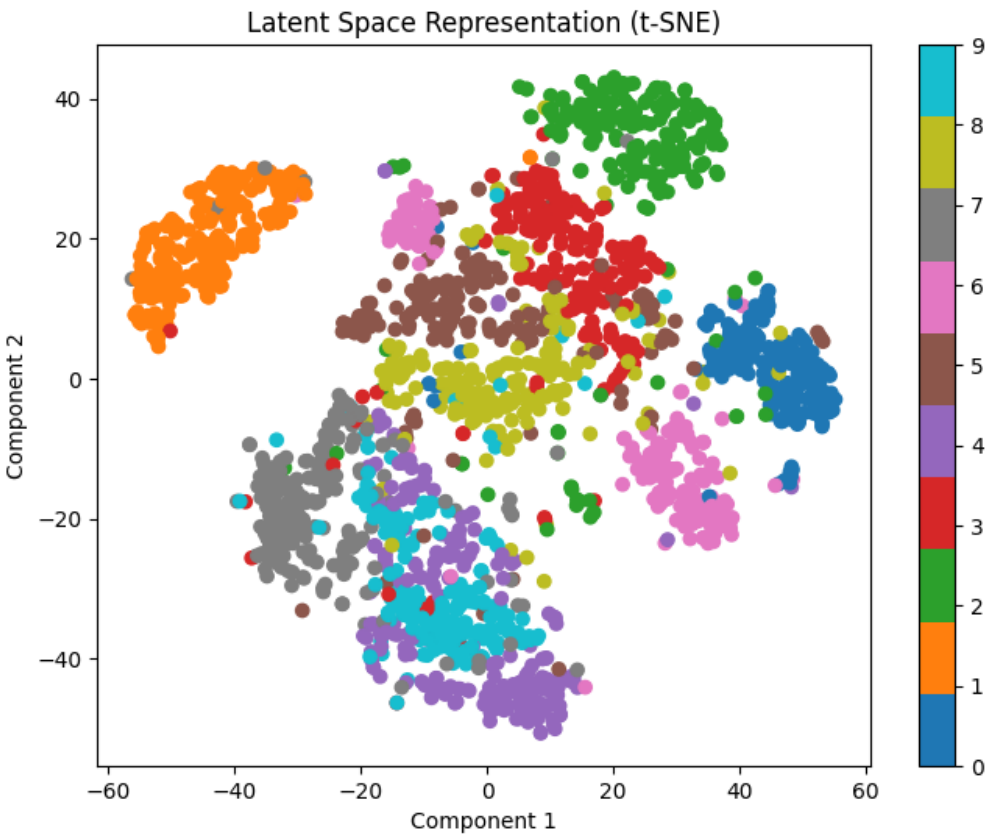
→**Reconstruction Performance:** The original and reconstructed images, as a comparison, depict that the autoencoder preserves the global structure of the digits. The images reconstructed, while minor details can be smoothened out, hold enough information for humans to recognize.

This shows there is a trade-off involved in the dimensionality reduction algorithm, some information is lost due to compression, but essential features are retained.



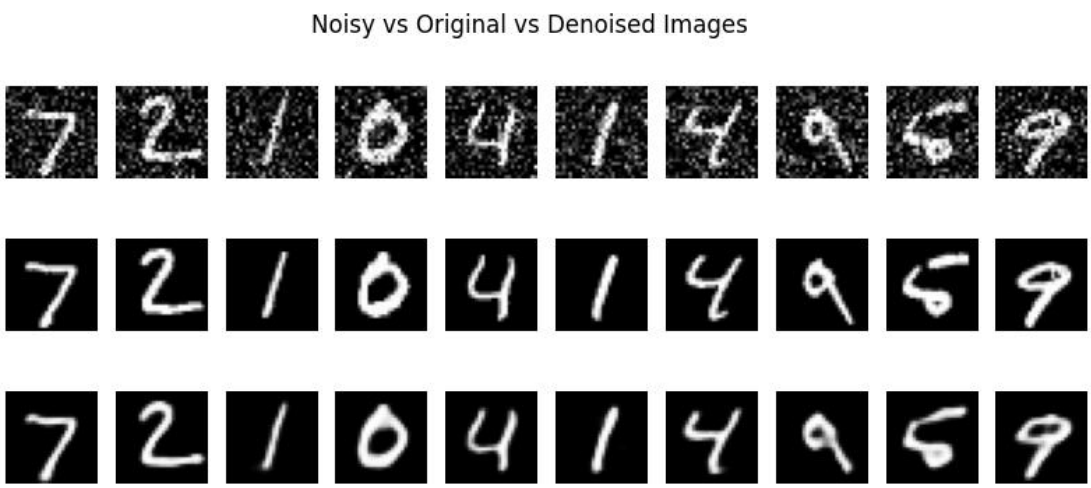
→**Structure of the Latent Space:** Clusters corresponding to digit classes can be visualized with latent space. Although the model is not trained on labels, it learns representations that separate the digits based on their visual characteristics.

By clustering an embedding, we know the autoencoder captures information beyond pixels.



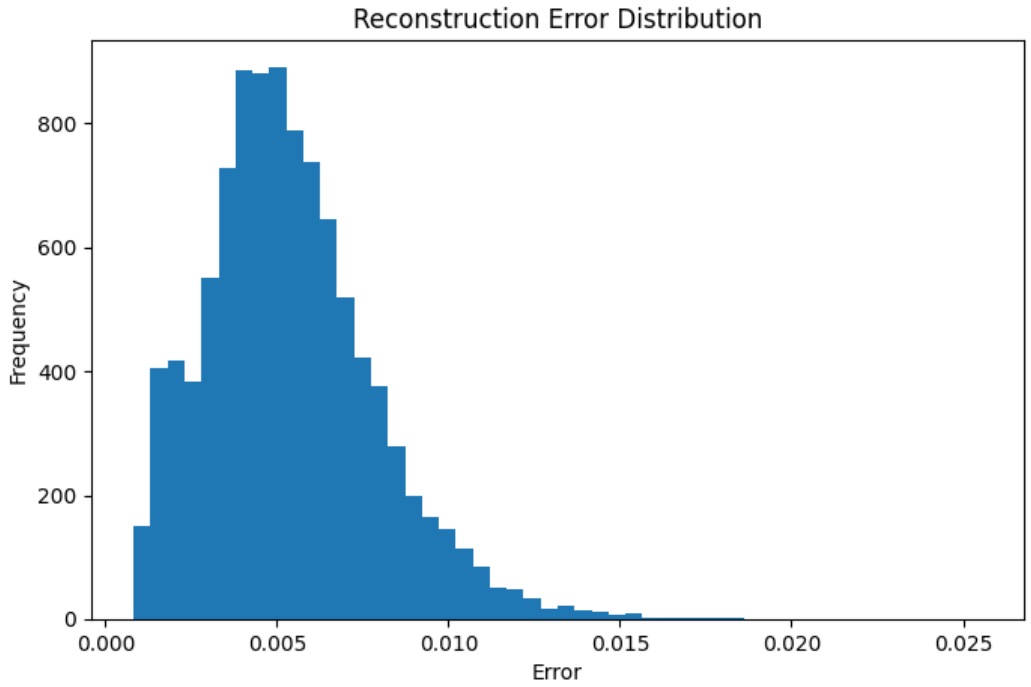
→**Impact of Denoising:** When noise is added to the input images this degrades the quality significantly. The denoising autoencoder is still able to reconstruct clean versions of these images accurately.

The model is learning the patterns instead of memorising the noisy inputs. So, it becomes more resilient and ready to operate on imperfect data it may encounter.



→**Reconstruction Error Analysis:** The distribution of reconstruction errors provides insight into model performance across different samples. Most images exhibit low error, indicating successful reconstruction. However, a small subset of images shows higher error values.

These high-error cases typically correspond to ambiguous or poorly written digits. This suggests that the model struggles with inputs that deviate significantly from common patterns.



Discussion:

The study's findings demonstrate the efficiency of autoencoders in achieving non-linear dimensionality reduction. The data redundancy is reduced in a latent space, where non-essential features are omitted.

The autoencoder can generate meaningful representations, and that too without any supervision which is a key observation. This clustering within the latent space suggests that the model is learning relationships between classes but is not explicitly empowered to classify them.

The denoising experiment reinforces the strong and powerful nature of the representations that were learned. The network aims to identify the fundamental structure of the data by forcing the model to reconstruct clean data from noisy inputs. It enhances generalisation and helps the model to be fit for a purpose.

Results however also show limitations. At times, the model fails on strange or ambiguous inputs, yielding a greater reconstruction error. This implies that autoencoders are efficient; however, efficiency significantly relies on the variety of the training data.

A key point is being able to interpret results. Although the latent space has meaningful structure components, we do not know what each dimension means. The learning of representation remains an open challenge.

In general, the experiments indicate that autoencoders can efficiently and flexibly accomplish dimensionality reduction tasks in domains with a non-linear relationship.

Limitations:

- Reconstruction quality depends heavily on architecture design.
- Some fine-grained details are lost during compression.
- Latent representations are not always easily interpretable.
- Performance may degrade on more complex datasets.
- t-SNE visualisation may distort true distances.

Conclusion:

This study employs convolutional autoencoders for downscaling image data to low dimensionality and recovering image data. The outcome suggested that autoencoders can export a compact interpretation of the input that has essential features.

- The encoder compresses arbitrary high-dimensional images into an information-rich lower-dimensional latent space.
- The reconstructive results show that the decoder successfully reconstructs the original structure of images.
- The visualisation of the latent space indicates the successful clustering of similar digits by the model.
- The experiment in denoising shows the robustness of the model in recovering clean images from noisy images.
- By analysing the reconstruction error, we gain insights on the limitations of the model and identify the more difficult samples.
- Autoencoders are far more flexible than traditional linear methods and can represent non-linear linkages.
- The global framework indicates that neural networks can serve not only for prediction but also for efficient data representation.

Future Directions of Work:

- Investigate Variational Autoencoders for probabilistic modelling.
- Apply the approach to more complex datasets such as CIFAR-10
- Combine autoencoders with classification models for hybrid systems.
- Explore deeper architectures for improved reconstruction quality.
- Use latent space representations for anomaly detection.

References

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MNIST Dataset: http://yann.lecun.com/exdb/mnist/